

Set Theory Refresher

A Self-Contained Document

1. Premise

Let the universal set be

$$U = \{0, 1, 2, \dots, 30\}$$

Define two subsets of U :

$$A = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

$$B = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$$

In words:

- **A** is the set of even numbers from 0 to 20.
- **B** is the set of multiples of 3 from 0 to 30.

All set operations below use this same universal set U , set A , and set B .

2. Core Set Vocabulary

- **Element:** $x \in A$ means x is in set A .
- **Not an element:** $x \notin A$ means x is not in set A .
- **Subset:** $A \subseteq B$ means every element of A is also in B .
- **Empty set:** $\emptyset$ is the set with no elements.
- **Cardinality:** $|A|$ is the number of elements in A .

Examples:

$$2 \in A, \quad 3 \in B, \quad 2 \notin B, \quad 3 \notin A$$

Neither $A \subseteq B$ nor $B \subseteq A$.

3. Cardinalities

$$|U| = 31$$

$$|A| = 11$$

$$|B| = 11$$

Inclusion–exclusion principle:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

4. Set Operations and Results

4.1 Union

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cup B = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 24, 27, 30\}$$

$$|A \cup B| = 18$$

4.2 Intersection

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

The common elements are multiples of both 2 and 3, i.e. multiples of 6:

$$A \cap B = \{0, 6, 12, 18\}$$

$$|A \cap B| = 4$$

4.3 Complement

$$A^c = U \setminus A$$

$$A^c = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$$

$$B^c = U \setminus B$$

$$B^c = \{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26, 28, 29\}$$

4.4 Set Difference

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

$$A \setminus B = \{2, 4, 8, 10, 14, 16, 20\}$$

$$B \setminus A = \{3, 9, 15, 21, 24, 27, 30\}$$

4.5 Symmetric Difference

$$A \Delta B = (A \setminus B) \cup (B \setminus A)$$

$$A \Delta B = \{2, 3, 4, 8, 9, 10, 14, 15, 16, 20, 21, 24, 27, 30\}$$

These are the elements in exactly one of A or B , but not both.

4.6 De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

For this example:

$$(A \cup B)^c = \{1, 5, 7, 11, 13, 17, 19, 22, 23, 25, 26, 28, 29\}$$

$(A \cap B)^c = \{1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$

5. Venn Diagram Summary

- **Only in A:** $\{2, 4, 8, 10, 14, 16, 20\}$
- **Only in B:** $\{3, 9, 15, 21, 24, 27, 30\}$
- **In both A and B:** $\{0, 6, 12, 18\}$
- **In neither A nor B:** $\{1, 5, 7, 11, 13, 17, 19, 22, 23, 25, 26, 28, 29\}$

6. Quick Reference Table

Operation	Notation	Result	Cardinality
Union	$A \cup B$	$\{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 24, 27, 30\}$	18
Intersection	$A \cap B$	$\{0, 6, 12, 18\}$	4
A minus B	$A \setminus B$	$\{2, 4, 8, 10, 14, 16, 20\}$	7
B minus A	$B \setminus A$	$\{3, 9, 15, 21, 24, 27, 30\}$	7
Symmetric difference	$A \Delta B$	$\{2, 3, 4, 8, 9, 10, 14, 15, 16, 20, 21, 24, 27, 30\}$	14
Complement of A	A^c	$\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$	20
Complement of B	B^c	$\{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26, 28, 29\}$	20
Complement of union	$(A \cup B)^c$	$\{1, 5, 7, 11, 13, 17, 19, 22, 23, 25, 26, 28, 29\}$	13
Complement of intersection	$(A \cap B)^c$	$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$	27

This document contains all necessary definitions, sets, operations, and results. No external references are required.